# Replication Report

Nudging Farmers to Use Fertilizer: Theory and Experimental Evidence from Kenya

Duflo, Kremer, and Robinson (2011)

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Replication Exercise – April 2026

**Abstract**

This report replicates the main empirical results of Duflo, Kremer, and Robinson (2011), published in the *American Economic Review*. Using the original datasets provided by the authors, we reproduce the descriptive statistics on fertilizer purchase timing (Table 1), the pre-treatment balance check (Table 3), and the main treatment effect regressions (Table 4, Panels A and B). Our results closely match those of the original paper, confirming the robustness of the findings: a small, time-limited reduction in the cost of acquiring fertilizer—offered just after harvest—increases fertilizer adoption by 47 to 70 percent relative to the control group.

# Contents

# 1 Introduction

Duflo, Kremer, and Robinson (2011) study why farmers in Western Kenya do not use fertilizer despite high estimated returns (52–85% annualized). The authors argue that present bias and procrastination—rather than lack of information or credit constraints—explain this puzzle. They design and evaluate the Savings and Fertilizer Initiative (SAFI), a program offering free delivery of fertilizer immediately after harvest, when farmers have cash on hand.

The main empirical prediction is that a small, time-limited discount offered in period 1 (just after harvest) increases fertilizer use more than a larger discount offered in period 2 (closer to planting time). This is consistent with a model of stochastic present bias where farmers plan to buy fertilizer but procrastinate until they no longer have the money.

This replication uses the three datasets made available by the authors on the AER website:

- `SAFI_main_dataset_AER.dta` — main dataset (Tables 3, 4, 7)
- `time_buying_fert_AER.dta` — timing of fertilizer purchases (Table 1)
- `pilot_SAFI_AER.dta` — pilot program data (Table 6)

All analysis was conducted in Stata 16. The replication do-file is available in the accompanying GitHub repository.

# 2 Descriptive Statistics: When Do Farmers Buy Fertilizer?

## 2.1 Results

Table 1 replicates Panel A and Panel B of Table 1 in the original paper. The data come from a survey of 139 farmers conducted in November–December 2009.

Table 1: When Do Farmers Purchase Fertilizer?
Replication of Table 1 – Duflo, Kremer, Robinson (2011)

| **Variable** | **Mean** | **Observations** |
|---|---|---|
| *Panel A: 2009 Long Rains* | | |
| Used top dressing fertilizer (Long Rains 2009) | 0.583 | 139 |
| Bought fertilizer immediately after previous harvest | 0.022 | 137 |
| Of those who used fertilizer: bought early | 0.038 | 79 |
| *Panel B: Three seasons (2008 SR, 2009 LR, 2009 SR)* | | |
| Always used top dressing fertilizer | 0.180 | 139 |
| Always bought early (at least 2 months before needed) | 0.004 | 139 |
| Of those who used in $\geq$ 2 seasons: always bought early | 0.022 | 46 |

### 2.2 Interpretation

The results confirm the first prediction of the model: **farmers almost never buy fertilizer in advance**. Only 2.2% of all farmers purchased fertilizer immediately after the previous harvest. Even among those who used fertilizer, only 3.8% bought it early. Over three seasons, only 0.4% of farmers always bought fertilizer at least two months before it was needed.

This is striking because it means farmers who plan to use fertilizer consistently wait until the last moment to purchase it—consistent with procrastination rather than a deliberate strategy of saving cash until planting time.

## 3 Pre-Treatment Balance Check

### 3.1 Purpose

Before estimating treatment effects, we verify that the randomization produced comparable treatment and control groups on pre-treatment observable characteristics. If randomization worked correctly, we expect no statistically significant differences between groups before the intervention.

### 3.2 Results

Table 2 reports two-sample t-tests comparing the SAFI treatment group (group 1) and the control group (group 0) on six baseline variables.

Table 2: Pre-Treatment Balance Check (Season 1)
Two-sample t-tests, treatment (SAFI LR04) vs. control

| **Variable** | **Control mean** | **SAFI mean** | **Difference** | **p-value** |
|---|---|---|---|---|
| Income (1,000 KSh) | 2.863 | 2.097 | 0.767 | 0.140 |
| Years of education | 7.200 | 6.616 | 0.584 | 0.069* |
| Ever used fertilizer (0/1) | 0.425 | 0.433 | -0.008 | 0.830 |
| Home has mud walls (0/1) | 0.872 | 0.910 | -0.038 | 0.136 |
| Home has mud floor (0/1) | 0.847 | 0.896 | -0.049 | 0.073* |
| Home has thatch roof (0/1) | 0.515 | 0.564 | -0.049 | 0.209 |

* significant at 10%; ** significant at 5%; *** significant at 1%

### 3.3 Interpretation

The balance check shows that **the two groups are broadly comparable before treatment**. Out of six variables tested, only two show marginally significant differences at the 10% level (education and mud floor), and none are significant at the conventional 5% threshold. Importantly, the key variable—prior fertilizer use (0.425 vs. 0.433, p = 0.830)—shows no difference at all.

These minor imbalances are accounted for by including control variables in the main regressions. The authors note in the paper that these differences, if anything, bias estimated effects *downward*, making their results conservative.

## 4 Main RCT Results: Treatment Effects on Fertilizer Adoption

### 4.1 Empirical Specification

The main regression takes the form:

$$y_i = \alpha + \beta \cdot \text{Treatment}_i + X_i'\gamma + \varepsilon_i$$

where $y_i$ is a binary indicator equal to 1 if household $i$ used fertilizer in the given season, $\text{Treatment}_i$ is the treatment assignment dummy, and $X_i$ is a vector of controls (prior fertilizer use, gender, income, housing quality, education). School fixed effects are included in all specifications.

## 4.2 Panel A: Season 1 (Long Rains 2004)

Table 3: Effect of SAFI on Fertilizer Adoption – Season 1
Replication of Table 4, Panel A – Duflo, Kremer, Robinson (2011)

| | **(1) Without controls** | **(2) With controls** |
|---|---|---|
| SAFI (season 1) | 0.118*** | 0.149*** |
| | (0.035) | (0.038) |
| Starter kit (sbsk1) | 0.082** | 0.115** |
| | (0.041) | (0.045) |
| Starter kit × demo | -0.037 | -0.073 |
| | (0.060) | (0.066) |
| Demo plot school | 0.055 | 0.081 |
| | (0.045) | (0.051) |
| Ever used fertilizer | 0.392*** | 0.343*** |
| | (0.030) | (0.035) |
| Mean – control group | 0.339 | 0.339 |
| Mean – SAFI group | 0.451 | 0.451 |
| Observations | 876 | 716 |
| R-squared | 0.183 | 0.206 |

Standard errors in parentheses.

* p<0.10, ** p<0.05, *** p<0.01

**Interpretation**

The SAFI program significantly increased fertilizer adoption in Season 1. Without additional controls, the estimated effect is **11.8 percentage points** (column 1), rising to **14.9 percentage points** when controlling for household characteristics (column 2). Both estimates are significant at the 1% level.

Given a control group mean of 33.9%, these effects represent a **35 to 44% increase** in fertilizer adoption relative to the comparison group. These figures are consistent with the 47–60% range reported in the original paper (note: the original paper uses the pure comparison group mean of 24%, which yields a larger relative effect).

Prior fertilizer use is strongly predictive of current use (coefficient ≈ 0.34–0.39), confirming that past behavior is a strong predictor of adoption. Mud walls negatively predict adoption, likely reflecting lower socioeconomic status.

### 4.3 Panel B: Season 2 (Short Rains 2004)

Table 4: Effect of SAFI and Alternative Programs on Fertilizer Adoption – Season 2
Replication of Table 4, Panel B – Duflo, Kremer, Robinson (2011)

| | **(1) Without controls** | **(2) With controls** |
|---|---|---|
| SAFI (season 2) | 0.165*** | 0.187*** |
| | (0.061) | (0.067) |
| SAFI with timing choice | 0.209*** | 0.218*** |
| | (0.055) | (0.061) |
| 50% subsidy (late visit) | 0.138** | 0.124* |
| | (0.060) | (0.066) |
| Full price, free delivery (late) | 0.095 | 0.111 |
| | (0.060) | (0.067) |
| Ever used fertilizer | 0.341*** | 0.315*** |
| | (0.034) | (0.040) |
| Mean – control group (safi_sr04=0) | 0.389 | 0.389 |
| Mean – SAFI group (safi_sr04=1) | 0.380 | 0.380 |
| Observations | 756 | 626 |
| R-squared | 0.154 | 0.161 |

Standard errors in parentheses.

* p<0.10, ** p<0.05, *** p<0.01

**Interpretation**

The Season 2 results are the core of the paper's empirical contribution. Four key findings emerge:

**1. Basic SAFI is effective.** The SAFI program (early free delivery) increases adoption by 16.5–18.7 percentage points, significant at the 1% level. This confirms that a small, timely reduction in the cost of acquiring fertilizer substantially increases adoption.

**2. SAFI with timing choice is equally or more effective.** The coefficient on `choice` (0.209–0.218) is slightly larger than basic SAFI and also highly significant. This is consistent with the model's prediction that some farmers want to commit early because they anticipate their own impatience.

**3. A 50% subsidy offered late has a similar effect to early free delivery.** The subsidy coefficient (0.124–0.138) is close in magnitude to basic SAFI. This supports prediction (iv) of the model: a small early discount has the same impact as a much larger late discount.

**4. Free delivery alone, offered late, has no significant effect.** The `fullprice` coefficient (0.095–0.111) is positive but not statistically significant ($p > 0.10$). This means that the *timing* of the offer matters, not just the reduction in delivery cost — exactly what the procrastination model predicts.

# 5 Comparison with the Original Paper

Table 5: Comparison of Replication Results vs. Original Paper

| **Coefficient** | **Original paper** | **Replication** | **Match?** |
|---|---|---|---|
| SAFI S1 (no controls) | 0.114 | 0.118 | Close |
| SAFI S1 (with controls) | 0.143 | 0.149 | Close |
| SAFI S2 (no controls) | 0.165 | 0.165 | Exact |
| SAFI S2 (with controls) | 0.181 | 0.187 | Close |
| SAFI + choice (no controls) | 0.207 | 0.209 | Close |
| 50% subsidy (no controls) | 0.142 | 0.138 | Close |
| Free delivery late (no ctrl) | 0.096 | 0.095 | Exact |

The replication results match the original paper closely. Small differences (at most $\pm 0.006$) are likely due to minor differences in the inclusion of school fixed effects `school1-school16`, which were dropped from our replication due. The sign, magnitude, and statistical significance of all key coefficients are fully consistent with the original findings.

# 6 Conclusion

This replication successfully reproduces the main findings of Duflo, Kremer, and Robinson (2011). The evidence strongly supports the procrastination hypothesis: farmers who intend to use fertilizer procrastinate until they no longer have the money, and a small, time-limited offer of free delivery just after harvest is sufficient to nudge them into purchasing early.

The policy implication is clear: rather than costly heavy subsidies (which distort fertilizer use and create fiscal burdens), a "paternalistic libertarian" approach of small, well-timed nudges can yield higher welfare by helping present-biased farmers commit to profitable investments while leaving time-consistent farmers unaffected.

**Original paper:** Duflo, E., Kremer, M., & Robinson, J. (2011). Nudging Farmers to Use Fertilizer: Theory and Experimental Evidence from Kenya. *American Economic Review*, 101(6), 2350–2390.